

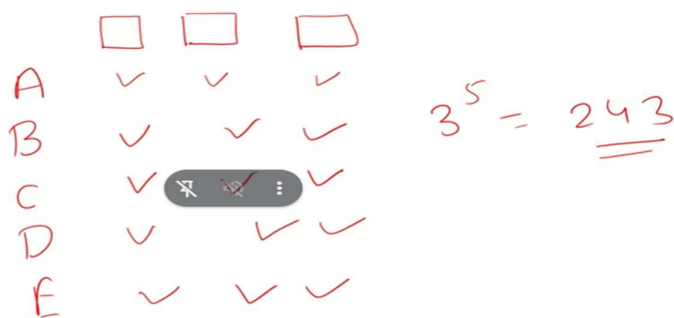
1) In how many ways 5 letters can be posted in 3 letter boxes available in the locality?

(a) 32

(b) 243

(c) 81

(d) 125



2) Find the number of ways in which 5 boys and 5 girls be seated in a row so that :

i. All the boys sit together and all the girls sit together.

ii. Boys and girls sit at alternate positions.

iii. No two girls sit together

$[B_1 B_2 B_3 B_4 B_5] [G_1 G_2 G_3 G_4 G_5]$

G B

$$5! \times 5! \times 2 = 120 \times 120 \times 2 = 28800$$

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$\frac{1}{1} B \frac{2}{2} B \frac{3}{3} B \frac{4}{4} B \frac{5}{5} B \frac{6}{6}$

$${}^6P_5 \times {}^5P_5$$

$$720 \times 120 =$$

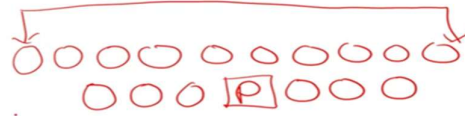
4) When a group photograph is taken, all the seven teachers should be in the first row and all the twenty students should be in the second row. If the two corners of the second row are reserved for the two tallest students, interchangeable only between them, and if the middle seat of the front row is reserved for the principal the number of such possible arrangements is

720 $\times$ 18!

1440 $\times$ 18!

360 $\times$ 18!

None of these



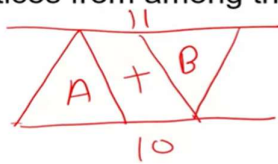
5) Ten points are marked on a straight line and eleven points are marked on another straight line. How many triangles can be constructed with vertices from among the above points?

495

550

1045

2475



$$[{}^{10}C_2 \times {}^{11}C_1] + [{}^{11}C_2 \times {}^{10}C_1]$$

$$45 \times 11 + 55 \times 10$$

$$495 + 550$$

$$\underline{1045}$$

6.

6) In how many different ways can letters of the word 'MISSISSIPPI' be arranged so that consonants always together?

A) 420

B) 840

C) 525

D) 635

$$1111 [msssspp]$$

$$12345$$

$$\frac{5! \times 7!}{4! \times 4! \times 2!}$$

$$= \frac{5 \times 7 \times 6 \times 5}{2} = \underline{\underline{525}}$$

7) How many numbers greater than 24000 can be formed by using the digits 1, 2, 3, 4, 5 when no digit is repeated ?

36

B) 60

C) 84

D) 120



8) Sam has forgotten his friend's seven-digit telephone number. He remembers the following:

1) The first three digits are either 635 or 674

2) The number is odd

3) The number 9 appears once.

$$\begin{aligned} 0 - 9 &= 10 \\ 0 - 8 &= 9 \checkmark \end{aligned}$$

What is the minimum number of trials he has to make before he can be certain to succeed.

A) 9999

B) 2430

C) 3401

D) 3006



$$2 \times 9 \times 9 \times 9 \times 1 =$$



Q9. From a pack of 52 cards, 4 cards were picked one at a time

A) If the card picked is not replaced, find the probability that all the cards were aces.

with Replacement

$$\frac{4}{52} \times \frac{4}{52} \times \frac{4}{52} \times \frac{4}{52}$$

$$\frac{1}{13^4}$$

without Replacement

$$\frac{4}{52} \times \frac{3}{51} \times \frac{2}{50} \times \frac{1}{49}$$

Q.10) In a box, there are 6 blue balls, X red balls and 10 green balls. Probability of choosing one red ball from the given box is $\frac{1}{3}$ then find the sum of red and blue balls in the box?

- a) 8
- b) 14
- c) 6
- d) 18

Q.12) A speaks truth in 60% cases and B speaks truth in 70% cases. Find the probability that they will say the opposite thing while describing a single event?

- (a) 0.44 (b) 0.54 (c) 0.46 (d) 0.70

Handwritten solution for Q.12:

	Truth	Lies
A	0.6	0.4
B	0.7	0.3

Diagram shows crossing lines between A's Truth and B's Lies, and A's Lies and B's Truth, indicating opposite statements.

Calculation: $\frac{60}{100} = 0.6$

$0.18 + 0.28 = 0.46$

More options for Shoab Attar

Q.13) Three 6 faces dice are thrown together find the probability that no two dice show the same number on them.

- a) $\frac{4}{9}$
- b) $\frac{5}{9}$ ✓
- c) $\frac{7}{8}$
- d) $\frac{4}{18}$

Handwritten solution for Q.13:

$6^1 = 6$

$6^2 = 36$

$6^3 = 216$

$\{1, 2, 3, 4, 5, 6\}$

Diagram showing three dice: $\square \square \square$

Calculation: $\frac{6 \times 5 \times 4}{216}$

$\frac{120}{216} = \frac{5}{9}$

36g

5/9

Q.14) What is the probability that a leap year selected at random will contain 53 Sundays?

a. $1/3$ (b) $1/7$ (c) $2/7$ (d) $1/14$ X

$$\frac{366 \text{ Days}}{7}$$

52 weeks 2 days
↑

52 Sundays

D_1

$$\frac{1}{7}$$

✓
X

D_2

$$+ \frac{1}{7} = \frac{2}{7}$$

X
✓

